2014-DSE
BIO
PAPER 1A

HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY
HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION 2014

BIOLOGY PAPER 1

8.30 am – 11.00 am (2 hours 30 minutes)
This paper must be answered in English

GENERAL INSTRUCTIONS

(1) There are **TWO** sections, A and B, in this Paper. You are advised to finish Section A in about 35 minutes.

(2) Section A consists of multiple-choice questions in this question paper. Section B contains conventional questions printed separately in Question-Answer Book **B**.

(3) Answers to Section A should be marked on the Multiple-choice Answer Sheet while answers to Section B should be written in the spaces provided in Question-Answer Book **B**. **The Answer Sheet for Section A and the Question-Answer Book B for Section B will be collected separately at the end of the examination.**

INSTRUCTIONS FOR SECTION A (MULTIPLE-CHOICE QUESTIONS)

(1) Read carefully the instructions on the Answer Sheet. After the announcement of the start of the examination, you should first stick a barcode label and insert the information required in the spaces provided. No extra time will be given for sticking on the barcode label after the 'Time is up' announcement.

(2) When told to open this book, you should check that all the questions are there. Look for the words '**END OF SECTION A**' after the last question.

(3) All questions carry equal marks.

(4) **ANSWER ALL QUESTIONS**. You are advised to use an HB pencil to mark all the answers on the Answer Sheet, so that wrong marks can be completely erased with a clean rubber. You must mark the answers clearly; otherwise you will lose marks if the answers cannot be captured.

(5) You should mark only **ONE** answer for each question. If you mark more than one answer, you will receive **NO MARKS** for that question.

(6) No marks will be deducted for wrong answers.

Not to be taken away before the end of the examination session

There are 36 questions in this section.
The diagrams in this section are NOT necessarily drawn to scale.

1. Which of the following is a catabolic process?

   A. Conversion of glucose to glycogen
   B. Absorption of glucose
   C. Emulsification of fat
   D. Digestion of starch

2. Which of the following events ***does not*** involve the functioning of membrane proteins?

   A. Transmission of nerve impulses across a synapse
   B. Absorption of glucose in the small intestine
   C. Transport of oxygen by haemoglobin
   D. Recognition of pathogens

Directions: Questions 3 to 5 refer to the following study:
A student wants to use an ordinary light microscope to observe the binary fission of a photosynthesizing protist under high magnification. A temporary mount of the protist is placed on the stage of the microscope.

3. Below are some steps in using a light microscope:

   (1) Focus with 10X objective
   (2) Focus with 40X objective
   (3) Search the field with 10X objective
   (4) Search the field with 40X objective
   (5) Move the slide until the protist is located in the centre of the field
   (6) Adjust light intensity if necessary

   Which of the following is the most reasonable sequence of steps for the above study?

   A. (1), (3), (5), (6)
   B. (2), (6), (4), (5)
   C. (1), (2), (4), (5), (6)
   D. (1), (3), (5), (2), (6)

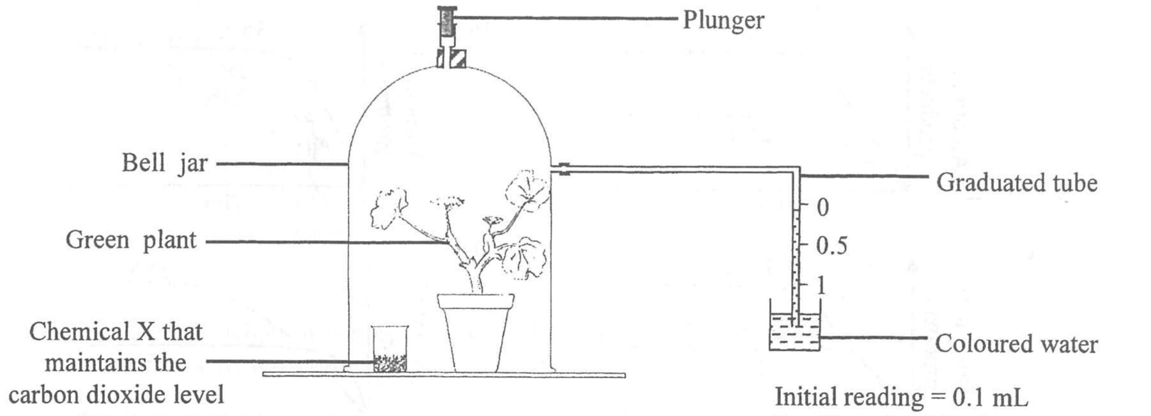
4. Which of the following correctly describes binary fission of the protist?

   A. The amount of organelles in the daughter cell is the same as that in the mother cell.
   B. The number of chromosomes in the daughter cell is half of that in the mother cell.
   C. The alleles found in the daughter cells are different from each other.
   D. The sizes of the daughter cells are similar to each other.

5. Which of the following structures would be observable in the above study?

   A. Ribosome
   B. Chloroplast
   C. Mitochondrion
   D. Endoplasmic reticulum

Directions: Questions 6 to 8 refer to the diagram below, which shows a set-up used to determine the rate of photosynthesis of a green plant. During the study, the position of the plunger remained unchanged.



Initial reading = 0.1 mL
Reading after 30 minutes = 0.4 mL

6. Based on the results, what was the rate of photosynthesis of this plant?

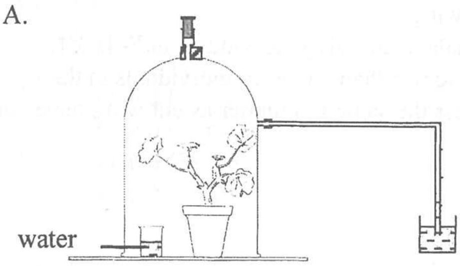
A. 0.6 mL oxygen released per hour
B. 0.3 mL oxygen released per hour
C. 0.6 mL carbon dioxide absorbed per hour
D. 0.3 mL carbon dioxide absorbed per hour

7. The rate obtained was lower than the actual rate of photosynthesis of the plant. Which of the following is the most probable reason for this?

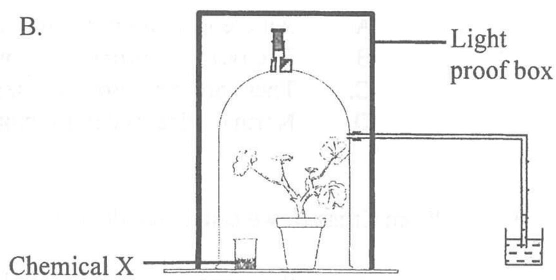
A. The plant also carried out respiration during the study.
B. The plant also carried out transpiration during the study.
C. The air temperature might have increased during the study.
D. The atmospheric pressure might have decreased during the study.

8. Which of the following set-ups can be used as a control for the above study to find out the actual rate of photosynthesis?

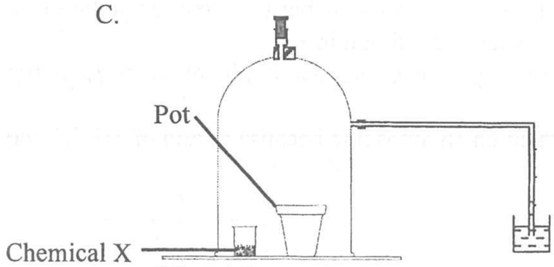
A.



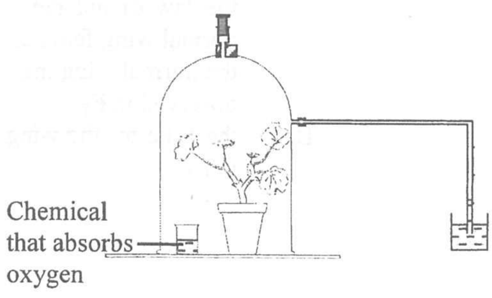
B.



C.

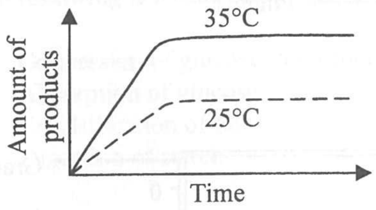


D.

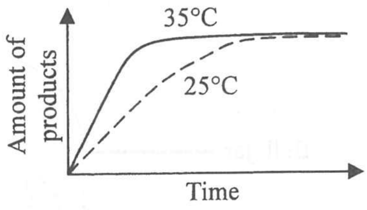


9. Which of the following graphs correctly shows the changes in the amount of products in a reaction catalyzed by a human enzyme at different temperatures?

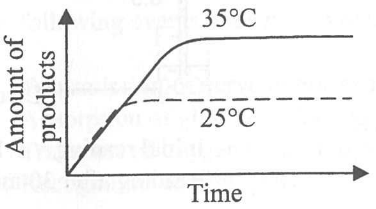
A.



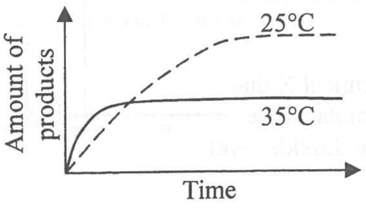
B.



C.



D.



Directions: Questions 10 and 11 refer to the following two crosses of fruit flies. In fruit flies, males are the heterogametic sex (XY) and the wing shape (normal wing or cut wing) is controlled by a single gene.

	Cross I	***Cross II***
Parents	Normal wing female x Cut wing male	Cut wing female x Normal wing male
F_1	12 normal wing females 11 normal wing males	11 normal wing females 11 cut wing males
F_2	71 normal wing females 34 normal wing males 35 cut wing males	32 normal wing females 33 cut wing females 36 normal wing males 38 cut wing males

10. Which of the following observations from Cross I best supports the conclusion that normal wing is the dominant phenotype?

A. All the F_1 individuals are normal wing.
B. The ratio of normal wing individuals to cut wing individuals in F_2 is 3:1.
C. There are more normal wing individuals than cut wing individuals in the F_2.
D. Normal wing males are more or less the same in number as cut wing males in the F_2.

11. From Cross II, we can conclude that

A. the gene for the wing shape is located on the X-chromosome because the cut wing phenotype was passed from the female parent to the F_1 males.
B. the law of independent assortment was demonstrated because new phenotypes, including normal wing females and cut wing males, were found in F_2.
C. the normal wing male parent is heterozygous because four combinations of phenotypes were observed in F_2.
D. the gene for the wing shape is located on an autosome because a ratio of 1:1:1:1 was shown in F_2.

12. Which of the following combinations of blood groups of parents may produce offspring with blood group AB?

(1) A x B
(2) AB x O
(3) AB x AB

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

13. The average height of men in a developed country rose by 10 cm between 1890 and 1980. Which of the following is the most probable reason for this observation?

A. Chemical pollution induced mutations leading to the increase in height.
B. Spontaneous mutation resulted in a shift to taller height.
C. The better nutrition supply promoted growth.
D. A taller height had a better chance of survival.

14. Which of the following processes produce ATP?

(1) Glycolysis
(2) Krebs cycle
(3) Conversion of pyruvate to lactic acid

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

15. Which of the organisms below belong to the domain Eukarya?

(1) Yeast
(2) Amoeba
(3) Mouse

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

16. Which of the following biomolecules are associated with transcription?

(1) DNA
(2) mRNA
(3) amino acid

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

Directions: Questions 17 and 18 refer to the following photographs of two different fish:

Fish X

Fish Y

17. Using the dichotomous key below to identify the fish:

1a	Both eyes on the top of the head	2
1b	One eye on each side of the head	3
2a	Has long whip-like tail	*Aetobatus narinari*
2b	Has short, blunt tail	*Bothus mancus*
3a	Has spots on its body surface	4
3b	Does not have spots on its body surface.................	5
4a	Has chin whiskers	*Pseudupeneus maculatus*
4b	Does not have chin whiskers	*Sphoeroides spengleri*
5a	Has stripes on its body surface	*Holocentrus rufus*
5b	Does not have stripes on its body surface	*Parapriacanthus guentheri*

	Fish X	***Fish Y***
A.	*Bothus mancus*	*Pseudupeneus maculatus*
B.	*Bothus mancus*	*Holocentrus rufus*
C.	*Aetobatus narinari*	*Parapriacanthus guentheri*
D.	*Aetobatus narinari*	*Sphoeroides spengleri*

18. Which of the following allow further study of the phylogenetic relationship between the two fish?

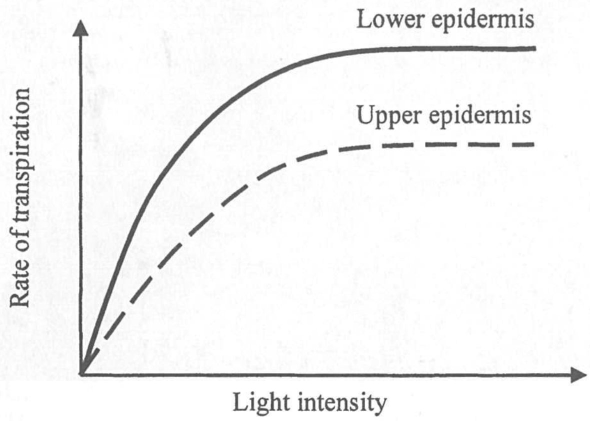
(1) Compare the amino acid sequences of their functional proteins
(2) Compare their internal body structure
(3) Compare their living habitats and behaviours

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

19. Which of the following is ***not*** an application of DNA fingerprinting?

A. Forensic science
B. Screening for genetic diseases
C. Sequencing of the human genome
D. Identification of Chinese medicines

Directions: Questions 20 and 21 refer to the graph below, which shows how the transpiration rates through the upper and lower epidermis of a leaf vary with light intensity:



20. Which of the following accounts for the difference in the transpiration rates through the upper and lower epidermis shown above?

    A. The mesophyll layer near the lower epidermis has more air spaces.
    B. The upper epidermis is more exposed to light.
    C. The air temperature below the leaf is lower.
    D. The upper epidermis has fewer stomata.

21. Which of the following methods can be used to measure the rate of transpiration through the upper and lower epidermis of the leaf?

    A. Peel the upper epidermis of the leaf and count the number of stomata under a microscope, and then repeat with the lower epidermis.
    B. Put the leaf into warm water and then count the number of bubbles that appear on each side of the leaf in a fixed period of time.
    C. Shine light on the upper epidermis of the leaf and measure the rate of water absorbed using a bubble potometer, and then repeat with the lower epidermis.
    D. Smear the upper epidermis of the leaf with vaseline and measure the rate of water loss using a weight potometer, and then repeat with the lower epidermis.

22. The transpiration rate of a tree is much higher than that of a herbaceous plant because the tree

    A. is much taller than the herbaceous plant.
    B. has many more roots than the herbaceous plant.
    C. has many more leaves than the herbaceous plant.
    D. has much more xylem than the herbaceous plant.

23. Which of the following cell types has the highest density of mitochondria?

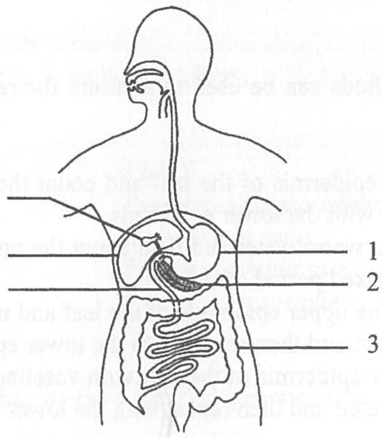
    A. Root hair cells
    B. Leaf epidermal cells
    C. Spongy mesophyll cells
    D. Palisade mesophyll cells

24. Which of the following dental formulae best represents the dentition shown in the photograph of the X-ray?

A. $\frac{2123}{2123}$ B. $\frac{2132}{2132}$

C. $\frac{3212}{3212}$ D. $\frac{2312}{2312}$

Directions: Questions 25 and 26 refer to the diagram below, which shows the human digestive system:



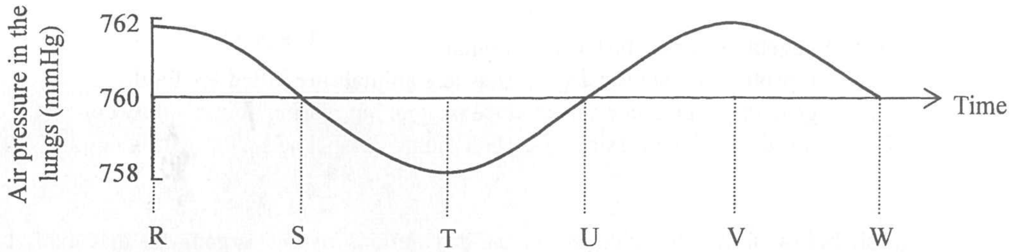
25. Physical digestion takes place at

A. 1 and 3.
B. 1 and 4.
C. 3 and 4.
D. 1, 3 and 4.

26. Which of the following structures are responsible for producing digestive juices that help the digestion of fat?

A. 2 and 5
B. 2 and 6
C. 5 and 6
D. 2, 5 and 6

27. The graph below shows the change in air pressure in the lungs of a man:

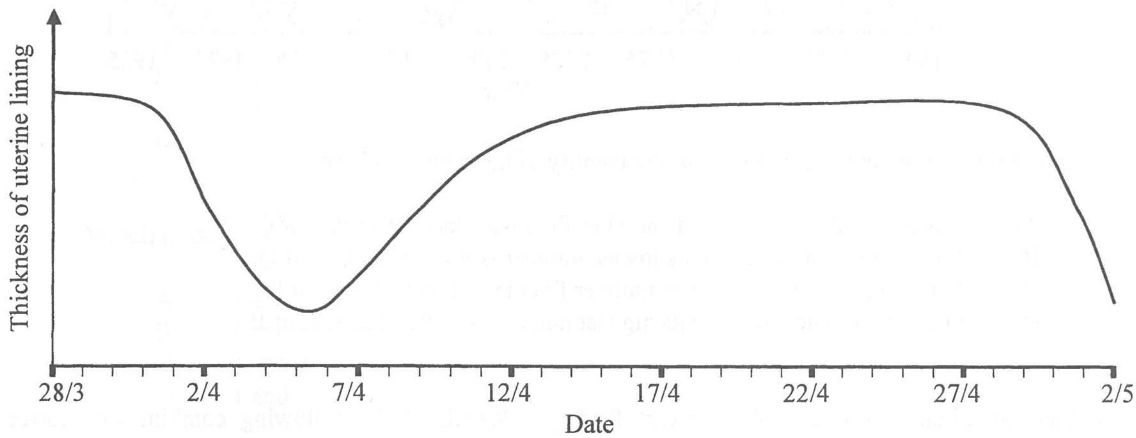


Atmospheric pressure = 760 mm Hg

His diaphragm muscles are in a state of contraction during the period

A. RT.
B. SU.
C. TV.
D. UW.

28. The diagram below shows the changes in the uterine lining of a woman:



During which of the following periods would the woman most likely get pregnant after copulation?

A. 28/3 to 1/4
B. 4/4 to 8/4
C. 11/4 to 15/4
D. 18/4 to 21/4

29. During foetal development, the placenta has functional roles similar to

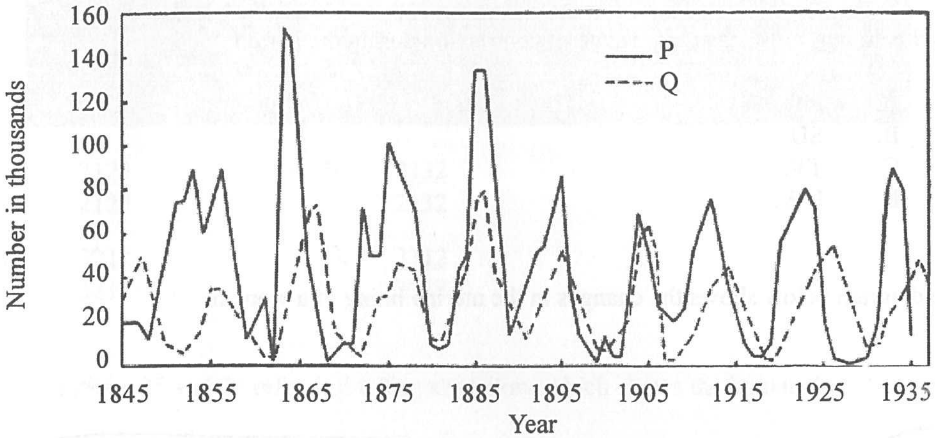
(1) the bone.
(2) the lungs.
(3) the small intestine.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

30. Some environmental protection groups claim that the vegetarian diet is good for our environment. This is probably because

A. vegetables grow faster than animals.
B. it protects endangered species as less animals are killed for food.
C. growing vegetables can produce oxygen but rearing animals only consumes oxygen.
D. it reduces the emission of carbon dioxide associated with rearing animals for food.

31. The graph below shows the changes in the populations of two organisms that exhibit a predator-prey relationship in a habitat:



Which of the following statements about the identity of the organism is correct?

A. P is the predator because its number fluctuates more than that of Q.
B. P is the predator because its lowest number is lower than that of Q.
C. Q is the predator because its number fluctuates less than that of P.
D. Q is the predator because its highest number is lower than that of P.

32. Infants can obtain antibodies from breast feeding. Which of the following combinations correctly describes this type of immunity in infants?

	Type of immunity	*Explanation*
A.	active	the antibodies are produced from white blood cells
B.	active	the antibodies attack pathogens bearing foreign antigens
C.	passive	the antibodies do not trigger the production of memory cells
D.	passive	the antibodies work only when there is re-entry of the same pathogen

33. Which of the following processes releases nitrogen-containing compounds from organisms back into the environment?

A. Nitrification
B. Decomposition
C. Denitrification
D. Nitrogen fixation

Directions: Questions 34 and 35 refer to Diagram I and Diagram II below. Diagram I shows a yoga instructor in a yoga posture. Diagram II shows some of the muscles associated with her left leg.

Diagram I

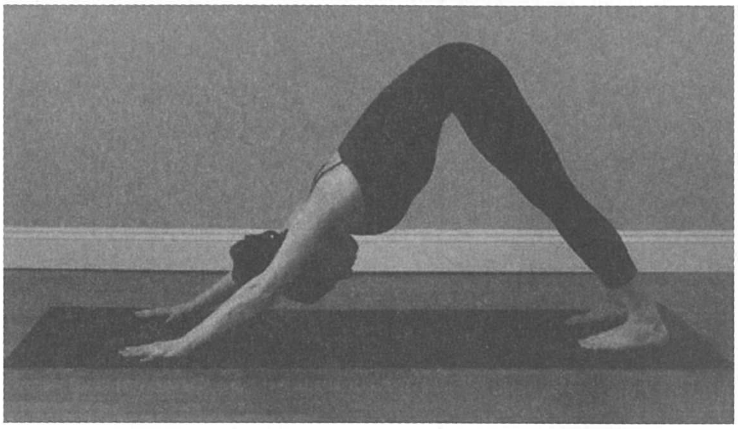
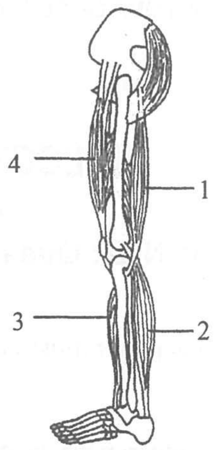


Diagram II



34. Which muscles of the left leg of the yoga instructor are contracting when she maintains the posture shown in diagram I?

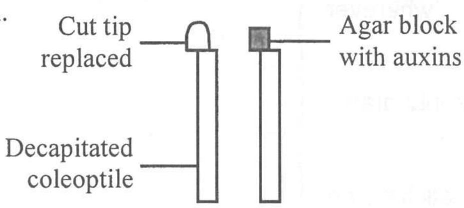
    A. 1 and 2
    B. 1 and 3
    C. 2 and 4
    D. 3 and 4

35. Which muscles of the left leg are flexors?

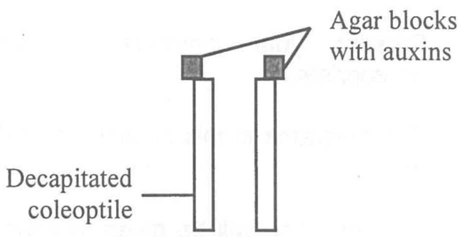
    A. 1 and 2
    B. 1 and 3
    C. 2 and 4
    D. 3 and 4

36. Which of the following pairs of set-ups can be used to test the hypothesis that auxins are growth-promoting substances in oat coleoptiles?

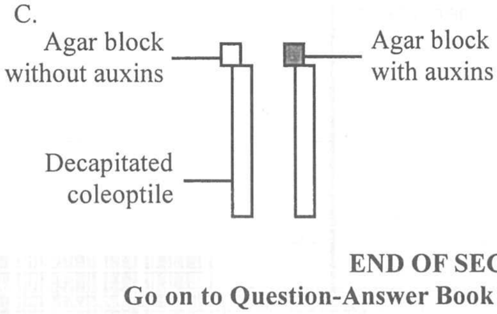
A.



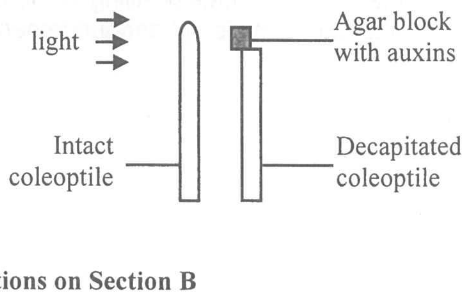
B.



C.



D.



END OF SECTION A

Go on to Question-Answer Book B for questions on Section B